# 7.2 The Central Limit Theorem

Now we come to the *very important* central limit theorem. First, let's introduce it intuitively as a process :

1. Suppose you have a large population (in theory infinite) with mean $\mu$ and standard deviation $\sigma$ (and any old shape).
2. Suppose you have a large sample, size $n$, of values from that population. (In practise we will see that $n > 30$ is large.) **Take the mean, $\overline{x}_1$, of that sample**. Put the sample back into the population[1]
3. Randomly pick another sample of size $n$. **Compute the mean of the new sample, $\overline{x}_2$**. Return the sample to the population..
4. Repeat step 3 an infinite number of times and **build up your collection of sample means $\overline{x}_i$**.
5. Then[2] the distribution of the sample means will be *normal* will have a mean equal to the population mean, $\mu$, and will have a standard deviation of

1. This is redundant since the population is infinite, but for conceptual purposes imagine that you return the items to the population.
2. More precisely, the distribution of sample means asymptotically approaches a normal distribution as $n \to \infty$. But 30 is close enough to infinity for most practical purposes and the statistical inferential tests that we will study will assume that the distribution of sample means will be normal.

$$\sigma_{\overline{x}} = \frac{\sigma}{\sqrt{n}}$$

where $\sigma$ is the population's standard deviation. $\sigma_{\overline{x}} = \sigma / \sqrt{n}$ is known as the *standard error of the mean.*

Now let's visualize this same process using pictures :

- Take a sample of size $n$ from the population and compute the mean $\overline{x}$ (see Figure 7.2a).

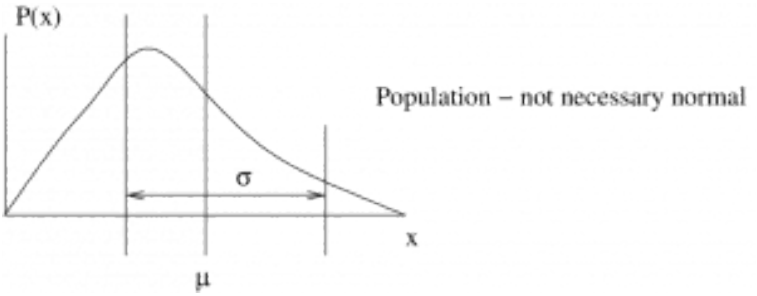



*Figure 7.2a*

- Put them back and take $n$ more data points.
- Do this over and over to get a bunch of values for $\overline{x}$. Those values for $\overline{x}$ will be distributed as shown in Figure 7.2b.

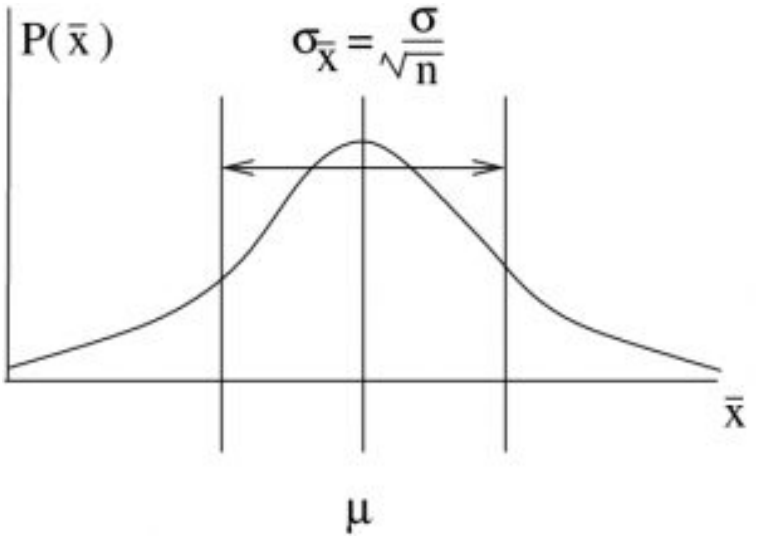



*Figure 7.2b*

The central limit theorem is our *fundamental sampling theory*. It tells us the *if* we know what the mean and standard deviation of a population[3] are *then* we can assign the probabilities of getting a certain mean $\overline{x}$ in a randomly selected sample from that population via a normal distribution of sample means that has the same mean as the population and a standard deviation equal to the standard error of the mean.

To apply this central limit theorem sampling theory we will need to compute areas $P$ under the normal distribution of means. In order to do that, so we can use the **Standard Normal Distribution Table**, we need to convert the values $(\overline{x})$ to a standard normal

3. In hypothesis testing we know what the mean of the population in the null hypothesis is.

$z$ using the $z$-tranformation as usual: $z = \frac{\overline{x}-\mu}{\sigma_{\overline{x}}}$. So, *for the distribution of sample means* the appropriate $z$-transformation is :

$$z = \frac{\overline{x} - \mu}{\sigma\sqrt{n}}$$

**Example 7.1** : Assume that we know, say from SGI's database, that the mean age of registered cars is $\mu = 96$ months and that the population standard deviation of the cars is $\sigma = 16$ months. We make no assumption about the shape of the population distribution. Then, what is the answer to the following sampling theory question: What is the probability that the mean age is between 90 and 100 months in a sample of 36 cars?

*Solution* : The central limit theorem tells us that sample means will be distributed as shown in Figure 7.3.

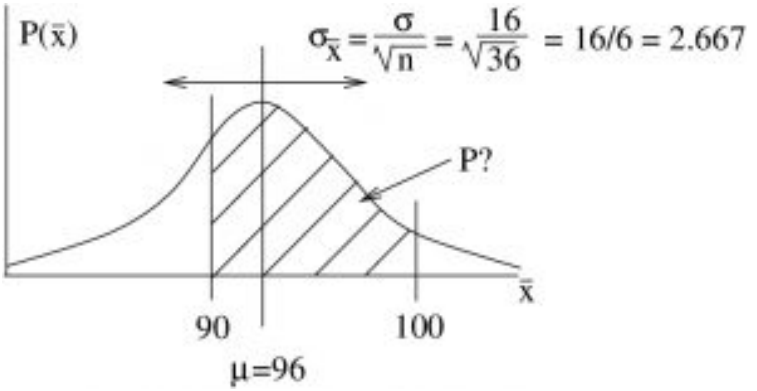



*Figure 7.3 : Distribution of mean age from samples of 36 cars.*

Convert 90 and 100 to $z$-scores as usual:

$$
\begin{aligned}
z(90) &= \frac{\bar{x}-\mu}{(\sigma/\sqrt{n})} = \frac{90-96}{2.667} = -2.25 \\
z(100) &= \frac{100-96}{2.667} = 1.50
\end{aligned}
$$

Then, the required probability using the **Standard Normal Distribution Table** is

$$
\begin{aligned}
P &= A(2.25) + A(1.50) \\
&= 0.4878 + 0.4332 \\
&= 0.921 \qquad (92.1\%)
\end{aligned}
$$

□

# 8. CONFIDENCE INTERVALS

# 8.1 Confidence Intervals Using the z-Distribution

With confidence intervals we will make our first statistical inference. Confidence intervals give us a direct inference about the population from a sample. The probability statement is one about hypotheses about the mean $\mu$ of the population based on the mean $\overline{x}$ and standard deviation $s$ of the sample. This is a fine point. The frequentist definition of probability gives no way to assign a probability to a hypothesis. How do you count hypotheses? The central limit theorem makes a statement about the sample means $\overline{x}$ on the basis of a hypothesis about a population, about its mean $\mu$ and standard deviation $\sigma$. If the population is fixed then the central limit theorem gives the results of counting sample means, frequentist probabilities. If we let $H$ represent a hypothesis about a population (i.e. that it is described by $\mu$ and $\sigma$) and let $D$ represent data (with mean $\overline{x}$) then the central limit theorem gives the probability $P(D \mid H) = P(\overline{x} \mid \mu, \sigma)$. The confidence intervals that we'll look at first give $P(H \mid D) = P(\mu \mid \overline{x}, \sigma)$. We'll look at the recipe for computing confidence intervals for means first, then return to this discussion about probabilities for hypotheses.

Our goal is to define a symmetric interval about the population mean $\mu$ that will contain all potentially measured values of $\bar{x}$ with a probability[1] of $\mathcal{C}$.

Typically $\mathcal{C}$ will be

1. Because of this issue about probabilities of hypotheses, many prefer to say "confidence" and not probability. But we will learn enough about Bayesian probability to say "probability".

$$\mathcal{C} = 0.90 \qquad (90\% \text{ confidence})$$
$$\mathcal{C} = 0.95 \qquad (95\% \text{ confidence})$$
$$\mathcal{C} = 0.99 \qquad (99\% \text{ confidence})$$

The assumptions that we need in order to use the $z$-distribution to compute confidence intervals for means are :

1. The population standard deviation, $\sigma$, is known (a somewhat artificial assumption since it is usually not known in an experimental situation) *or*
2. The sample size is greater than (or equal to) 30, $n \geq 30$ and we use $\sigma = s$, the sample standard deviation in our confidence interval formula.

*Definition* : Let $z_{\mathcal{C}} = z_{\alpha/2}$ where $\mathcal{C} = 1 - \alpha$ be the $z$-value, from the **Standard Normal Distribution Table** that corresponds to an area, between 0 and $z_{\mathcal{C}}$ of $\mathcal{C}/2$ as shown in Figure 8.1.

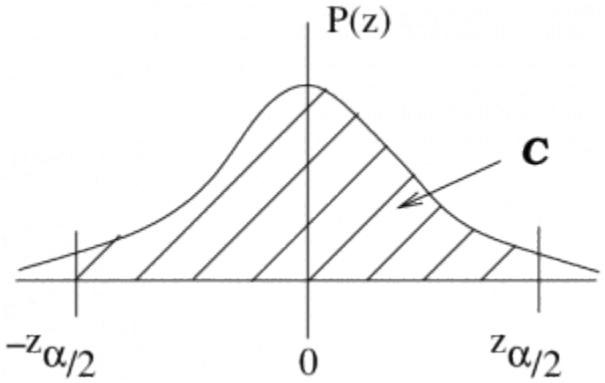



*Figure 8.1 : The $z$-distribution areas of interest associated with $z_{\mathcal{C}} = z_{\alpha/2}$.*

To get our confidence interval we simply inverse $z$-transform the picture of Figure 8.1, taking the mean of 0 to the sample mean $\bar{x}$ and

the standard deviation of 1 to the standard error $\sigma/\sqrt{n}$ as shown in Figure 8.2.

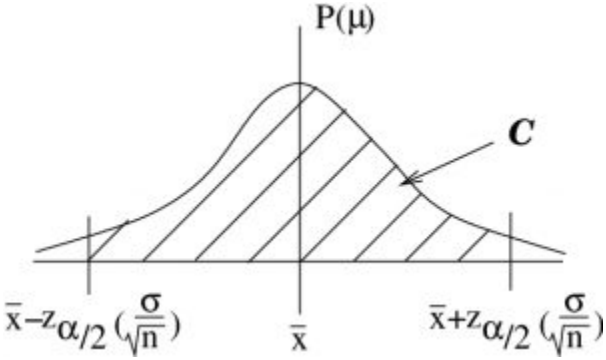



*Figure 8.2 : The inverse $z$-transformation of Figure 8.1 gives the confidence interval for $\mu$.*

So here is our recipe from Figure 8.2. The $\mathcal{C}$-confidence interval for the mean, under one of the two assumptions given above, is :

$$\bar{x} - z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right) < \mu < \bar{x} + z_{\alpha/2}\left(\frac{\sigma}{\sqrt{n}}\right)$$

or using notation that we will use as a standard way of denoting symmetric confidence intervals

(8.1) $\bar{x} - E < \mu < \bar{x} + E$

where

$$E = z_{\mathcal{C}}\left(\frac{\sigma}{\sqrt{n}}\right).$$

The notation $z_{\mathcal{C}}$ is more convenient for us than $z_{\alpha/2}$ because we will use the **t Distribution Table** in the Appendix to find $z_{\mathcal{C}}$ very quickly. We could equally well write

$$\mu = \bar{x} \pm E$$

but we will use Equation (8.1) because it explicitly gives the bounds for the confidence interval.

Notice how the confidence interval is *backwards* from the picture that the central limit theorem gives, the picture shown in Figure 8.3. We actually had no business using the inverse $z$-transformation $\mu = (z - \bar{x})/(\sigma/\sqrt{n})$ to arrive at Figure 8.2. It reverses the roles of $\mu$ and $\bar{x}$. We'll return to this point after we work through the mechanics of an example.

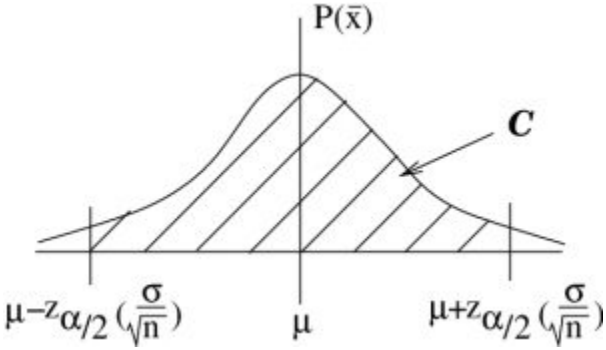



*Figure 8.3 : The central limit theorem is about distributions of sample means.*

**Example 8.2** : What is the $95\%$ confidence interval for student age if the population $\sigma$ is 2 years, sample $n = 50, \bar{x} = 23.2$?

*Solution* : So $\mathcal{C} = 0.95$. First write down the formula prescription so you can see with numbers you need:

$$\bar{x} - E < \mu < \bar{x} + E \qquad \text{where} \qquad E = z_{95\%}\frac{\sigma}{\sqrt{n}}.$$

First determine $z_{\mathcal{C}} = z_{\alpha/2}$. With the tables in the Appendices, there are two ways to do this. The first way is to use the **Standard Normal Distribution Table** noting that we need the $z$ associated with a table area of $0.95/2 = 0.475$. Using the table backwards we find $z_{\mathcal{C}} = 1.96$. The **second way**, *the recommended way*

especially during exams, is to use the **t Distribution Table**. Simply find the column for the $95\%$ confidence level and read the $z$ from the last line of the table. We quickly find $z_{95\%} = 1.960$.

Either way we now find

$$E = 1.96(\frac{2}{\sqrt{50}}) = 0.6$$

so

$$\begin{aligned} \bar{x} - E &< \mu < \bar{x} + E \\ 23.2 - 0.6 &< \mu < 23.2 + 0.6 \\ 22.6 &< \mu < 23.8 \end{aligned}$$

with $95\%$ confidence.

□

# 8.2 **Bayesian Statistics

Now that we've seen how easy it is to compute confidence intervals, let's give it a proper probabilistic meaning. To extend probability from the frequentist definition to the Bayesian definition, we need *Bayes' rule*. Bayes' rule is, for events $A$ and $B$ :

$$P(A \mid B)P(B) = P(B \mid A)P(A).$$

Study Figure 8.4 to convince yourself that Bayes' rule is true. Notice that

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

and

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}.$$

So, equating $P(A \cap B)$ from each of those two perspectives, we get Bayes' rule.

If we let $A = H$ (hypothesis) and $B = D$ (data), Bayes' rule gives us a way to define the probability of hypothesis through

$$P(H \mid D) = P(D \mid H) \left[ \frac{P(H)}{P(D)} \right]. \tag{8.2}$$

The quantity $[P(H)/P(D)]$ is known as the *prior probability* of the data relative to the hypothesis and is something that can be computed in theory if probabilities are assigned in a reasonable manner. The specification of prior probabilities is a contentious issue with the Bayesian approach. Really, it represents a prior *belief*. The quantity $P(D \mid H)$ is what sampling theory, like the central limit theorem, gives and is known as the *likelihood*. Finally the quantity $P(H \mid D)$ is known as the *posterior probability*. Equation (8.2) is an expression about probability

distributions as well as individual probabilities (just allow $H$ and $D$ to vary).

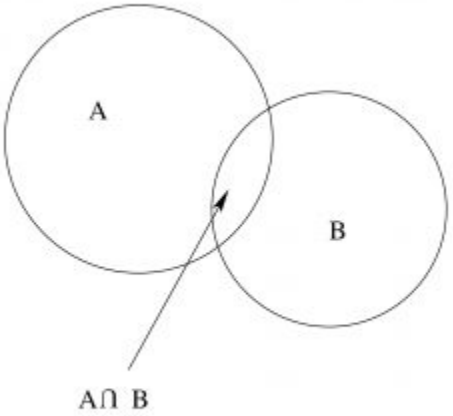



*Figure 8.4 : Venn diagram illustration of Bayes rule.*

If we assign $[P(H)/P(D)] = 1$ for the prior probability then $P(H \mid D) = P(D \mid H)$. We can switch the roles of $D$ and $H$! Of course $[P(H)/P(D)] = 1$ is not a probability distribution because the area under a function whose value is always 1 is infinite. The area under a probability distribution must be 1. So $[P(H)/P(D)] = 1$ is an *improper distribution* (as a function of either $H$ or $D$). But note that an improper distribution times a proper distribution here gives rise to a proper distribution. With this slight of hand, we can give confidence intervals a probabilistic interpretation.

# 8.3 The t-Distributions

As a broad introduction, the $t$-distributions are family of distributions that give different approximations to the $z$ -distribution as shown in Figure 8.5.

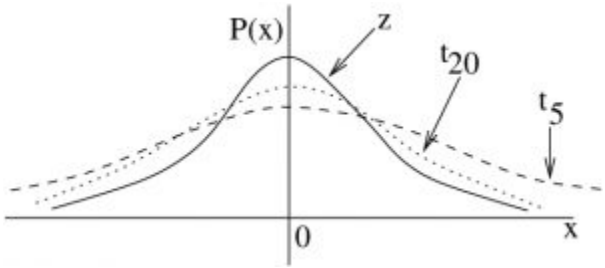



*Figure 8.5 : The $t$-distributions are a family of distributions, labeled here by their degrees of freedom $\nu$ as in $t_\nu$.*

As the degrees of freedom, $\nu$, increases, $t_\nu$ become closer to $z$, $\lim_{\nu\to\infty} t_\nu = z$. In practice, as reflected in the **t Distribution Table**, $t_{30}$ is very very close to $z$.

The $t$-distributions arise as a corollary to the central limit theorem; they give the distribution of sample means when knowledge of the population $\sigma$ is replaced by using the sample mean $s$. When we encounter the $\chi^2$ distribution later, we will give a more exact mathematical specification of the $t$-distributions.

Similar, to the $z$-distribution case, the $\mathcal{C}$ confidence interval for the mean $\mu$ for small $n$ samples is given by

$$\bar{x} - E < \mu < \bar{x} + E$$

where, now

$$E = t_{\nu,\mathcal{C}} \left( \frac{s}{\sqrt{n}} \right).$$

With this new formula for $E$ we have replaced $\sigma$ with $s$ in

comparison with the formula we used in Section 8.1: Confidence Intervals using the z-distribution and, of course, replaced $z_{\mathcal{C}}$ with $t_{\nu,\mathcal{C}}$. Some books use $t_{\nu,\mathcal{C}} = t_{\nu,\alpha/2}$ like the $z_{\mathcal{C}}$ of Section 8.1. We use $t_{\nu,\mathcal{C}}$ because we'll look up its value in the **t Distribution Table** in the column for $\mathcal{C}$ confidence intervals (just like we did with $z$) and with the degrees of freedom $\nu$ specifying the row. The formula for the degrees of freedom in this case is :

$$\nu = n - 1.$$

The $t_{\nu,\mathcal{C}}$ specify a probability $\mathcal{C}$ as shown in Figure 8.6. As before, the inverse $z$-transform, in the form $x = t_{\nu,\mathcal{C}} s + \bar{x}$ from the $t$-distribution on the left of Figure 8.6 to the distribution on the right of Figure 8.6 leads to our confidence interval formula for small means. And as before we should justify using that transform from a Bayesian perspective.

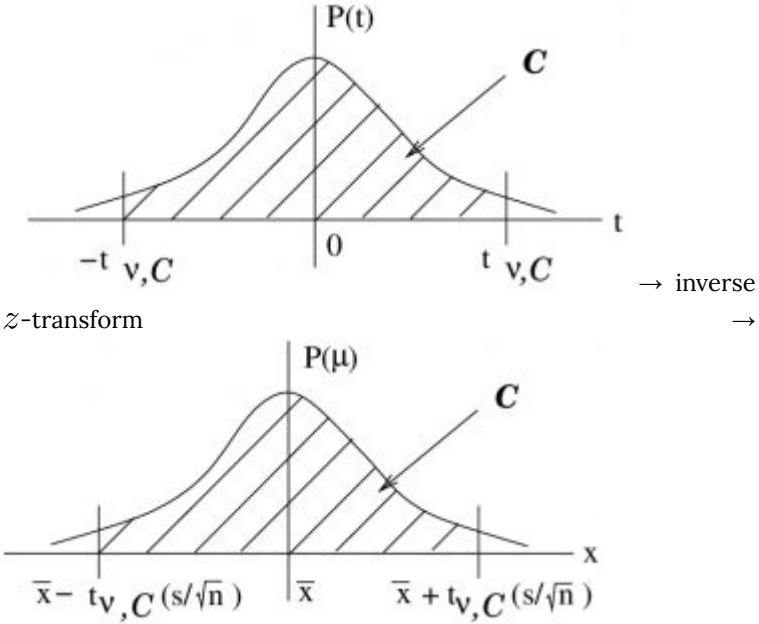



*Figure 8.6 : Derivation of confidence intervals for means of small samples.*

**Example 8.2** : Given the following data:

5460 5900 6090 6310 7160 8440 9930

find the 99% confidence interval for the mean.

*Solution* : First count $n = 7$ and then, with your stats calculator compute

$$\bar{x} = 7041.4 \quad \text{and} \quad s = 1610.3.$$

Using the **t Distribution Table** with $\nu = n - 1 = 6$ in the 99% confidence interval column, find

$$t_{n-1,\mathcal{C}} = t_{6,99\%} = 3.707.$$

With these numbers, compute

$$E = t_{n-1,\mathcal{C}} \left( \frac{s}{\sqrt{n}} \right) = 3.707 \left( \frac{1610.3}{\sqrt{7}} \right) = 2256.2$$

so

$$\begin{aligned} \bar{x} - E &< \mu < \bar{x} + E \\ 7041.4 - 2256.2 &< \mu < 7041.4 + 2256.2 \\ 4785.2 &< \mu < 9297.6 \end{aligned}$$

is the $99\%$ confidence interval for $\mu$.

□

# 8.4 Proportions and Confidence Intervals for Proportions

We will now make use of the approximation of the binomial distribution by the $z$-distribution given in Section 7.1: Using the Normal Distribution to Approximate the Binomial Distribution. As usual, the confidence interval will switch the roles of population and sample quantities. The recipe will be laid out first, then we will connect it to what you know about the binomial distribution.

First some definitions. Let $X$ be the number of items in a population of size $N$ that have a given quality. (e.g. the number of females in a population; or the number of people at the U of S wearing yellow sweaters). Then the proportion of the population having the given quality is

$$p = \frac{X}{N}$$

Given a sample from the population of size $n$, the best estimate for $p$ is:

$$\hat{p} = \frac{x}{n}$$

where $x$ is the number of items in the sample having the given quality. To go along with $\hat{p}$ we also have

$$\hat{q} = 1 - \hat{p}$$

which is is the proportion of items in the sample without the given quality.

To compute an $\mathcal{C}$ confidence interval for a proportion $p$ we need to compute

$$E = z_{\mathcal{C}}\sqrt{\frac{\hat{p}\hat{q}}{n}}$$

and it must be true that both $n\hat{p} \geq 5$ and $n\hat{q} \geq 5$ (otherwise we need to use the binomial distribution directly).

With $E$, the $\mathcal{C}$ confidence interval for a proportion is given by

$$\hat{p} - E < p < \hat{p} + E.$$

To derive the proportions confidence interval formula we'll begin with the sampling theory given by the binomial distribution and the corresponding $z$-approximation. Then we'll switch the roles of $p$ and $\hat{p}$. Let

$$x_{\text{pop}} = \frac{n}{N}X = np$$

be the mean, the expected value, of $x$ that you expect to find in a sample of size $n$ randomly selected from the population with a proportion $p$ of items of interest. This is true because $p$ is also the probability of randomly selecting an item of interest (the probability of success) from the population as per what we did in Chapter 4. The binomial distribution tells you the probability of getting different numbers $x$ of items of interest in your sample given $p$. The binomial distribution that describes our situation is shown in Figure 8.7; it has a standard deviation of $\sigma = \sqrt{npq}$.

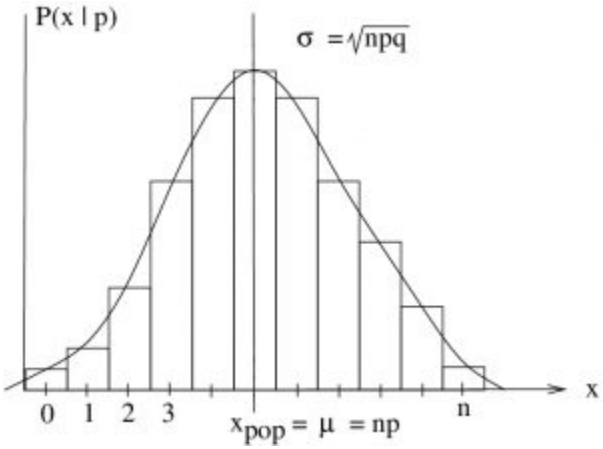



*Figure 8.7 : The binomial distribution relevant to forming a sample of size $n$ with $x$ items of interest from a population with a proportion $p$ of items of interest. The normal distribution with the same $\mu$ and $\sigma$ is shown.*

Moving to the normal approximation, we have the picture of Figure 8.8.

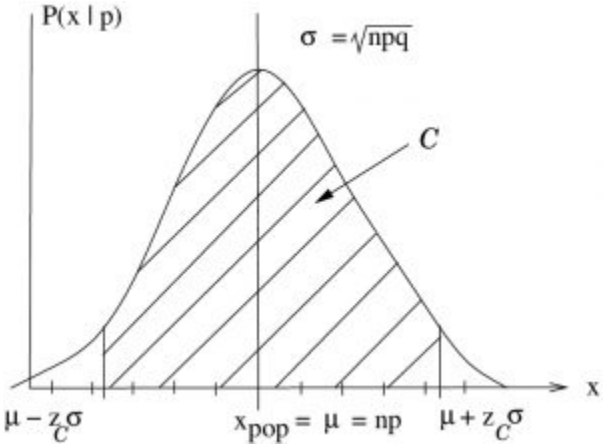



*Figure 8.8 : The normal distribution relevant to forming a sample of size $n$ with $x$ items of interest from a population with a proportion $p$ of items of interest. The boundaries of the area $C$ follow from an inverse $z$-transform of the $z$ -distribution to a normal distribution of mean $\mu$ and standard deviation $\sigma$, $x = z\sigma + \mu$.*

Figure 8.8 says :

$$\begin{aligned} \mu - z_C\sigma &< x < \mu + z_C\sigma \\ np - z_C\sqrt{npq} &< x < np + z_C\sqrt{npq} \end{aligned}$$

with a (frequentist) probability of $C$. This is our sampling theory. Divide by $n$:

$$\begin{aligned} p - z_C\sqrt{\frac{pq}{n}} &< \frac{x}{n} < p + z_C\sqrt{\frac{pq}{n}} \\ p - z_C\sqrt{\frac{pq}{n}} &< \hat{p} < p + z_C\sqrt{\frac{pq}{n}} \end{aligned}$$

Swapping the roles of the population and sample, we arrive at the confidence interval formula :

$$\hat{p} - z_{\mathcal{C}}\sqrt{\frac{\hat{p}\hat{q}}{n}} \;<\; p \;<\; \hat{p} + z_{\mathcal{C}}\sqrt{\frac{\hat{p}\hat{q}}{n}}.$$

Time for a worked example.

**Example 8.3** : A sample of 500 nursing applications included 60 men. Find the 90% confidence interval of the true proportion of men who applied to the nursing program.

*Solution* : From the ***t* Distribution Table**, look up

$$z_{\mathcal{C}} = 1.65$$

and compute

$$\hat{p} = \frac{x}{n} = \frac{60}{500} = 0.12$$

$$\hat{q} = 1 - \hat{p} = 1 - 0.12 = 0.88$$

$$E = z_{\alpha/2} \cdot \sqrt{\frac{\hat{p}\hat{q}}{n}} = 1.65\sqrt{\frac{(0.12)\cdot(0.88)}{500}} = 0.024.$$

Then

$$\hat{p} + E < p < \hat{p} - E$$
$$0.12 - 0.024 < p < 0.12 + 0.024$$
$$0.096 < p < 0.144$$

is the confidence interval with 90% confidence.

□

**Sample size need for a poll**

Measuring proportions is what pollsters do. For example in an election you might want to know how many people will vote for liberals (items of interest) and how many will vote for conservatives (items not of interest)[1] In a news paper you might see: "The poll

1. We assume here that there are only two parties. For the real life situation of more than two parties we need the

says that $72\%$ of the voters will vote liberal. The poll is considered accurate to 2 percentage points 19 time out of 20." This means that the $95\%$ confidence interval (19/20 = 0.95) of the proportion of liberal voters is $0.72 \pm 0.02$ (note how proportions are presented as percentages in the newspaper). The error here is $E = 0.02$. Before the pollster starts telephoning people, she must know how many people to phone to arrive at that goal error of $2\%$. She needs to know what the sample size $n$ needed is. In general, the minimum sample size needed to attain a goal error $E$ on a confidence interval of $C$ is

$$n = \hat{p}\hat{q}\left(\frac{z_C}{E}\right)^2.$$

Here $\hat{p}$ and $\hat{q}$ could come from a previous survey if available. If there is no such survey or if you want to be sure of ending up with an error equal to or less than a goal E, then use $\hat{p} = \hat{q} = 0.5$, see Figure 8.9.

multinomial distribution and to approximate it with a multivariate normal distribution. That is a topic for multivariate statistics but the principles are the same as what we cover here.

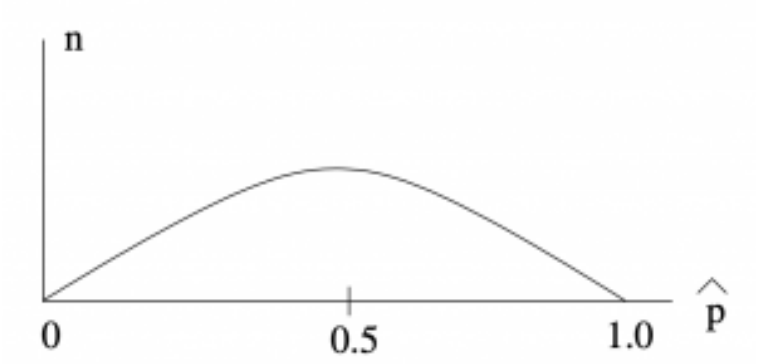



*Figure 8.9 : The formula* $n = \hat{p}\hat{q}\left(\frac{z_{\mathcal{C}}}{E}\right)^2$ *is a quadratic formula. Substitute* $\hat{q} = 1 - \hat{p}$ *to get* $n = \hat{p}(1-\hat{p})\left(\frac{z_{\mathcal{C}}}{E}\right)^2$ *or* $n = (\hat{p} - \hat{p}^2)\left(\frac{z_{\mathcal{C}}}{E}\right)^2$. *The maximum of* $n_{\max} = \frac{1}{4}\left(\frac{z_{\mathcal{C}}}{E}\right)^2$ *is at* $\hat{p} = 0.5$.

**Example 8.4** : We want to estimate, with $95\%$ confidence, the proportion of people who own a home computer. A previous study gave an answer of $40\%$. For a new study we want an error of $2\%$. How many people should we poll?

*Solution* : From the question we have :

$$\hat{p} = 0.40, \qquad \hat{q} = 0.60$$
$$E = 0.02, \qquad \alpha = 0.95$$

From the **t Distribution Table** (or the **Standard Normal Distribution Table** if you think about the areas correctly) we find

$$z_{\mathcal{C}} = z_{95\%} = 1.960.$$

Therefore

$$n = \hat{p}\hat{q}\left(\frac{z_{\alpha/2}}{E}\right)^2 = (0.40)(0.60)\left(\frac{1.96}{0.02}\right)^2 = 2304.96$$

Which we round up to a sample size of 2305 to ensure that $E < 0.02$.

□

# 8.5 Chi Squared Distribution

The $\chi^2$ (chi squared) distribution is a consequence of a random process based on the normal distribution. It is derived from the normal distribution as the result of the following stochastic process :

1. Suppose you have a population that has variance $\sigma^2$ and is normally distributed.
2. Take a sample of size $n$ from the population and compute $x_1 = \frac{(n-1)s_1^2}{\sigma^2}$ using the sample standard deviation $s_1$ from that sample.
3. Put the sample back into the population.
4. Take another sample of size $n$ from the population and compute $x_2 = \frac{(n-1)s_2^2}{\sigma}$ using the sample standard deviation $s_2$ from that sample.
5. etc.
6. The distribution of the values of $x_i = \frac{(n-1)s_i^2}{\sigma^2}$ values will be a $\chi^2$ distribution with $\nu = n - 1$ degrees of freedom.

Like the $t$-distributions, the $\chi^2$ distributions are a family, see Figure 8.10.

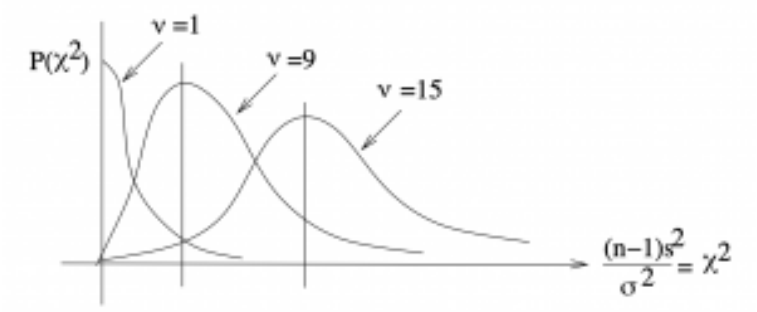



*Figure 8.10 : The $\chi^2$ distributions are enumerated by degrees of freedom.*

The $\chi^2$ distribution underlies why $s$ is the best estimate for $\sigma$. It mean, or expected value is $\nu = n - 1$ so the expected value of $s$ is $\sigma$. The expected value of $\sum(x - \bar{x})/n$ in a random sample of size $n$ is not $\sigma$.

**Confidence Intervals on $\sigma$ and $\sigma^2$**

The $\chi^2$ distribution is already normalized in its definition through including $s$ in its definition. Therefore no $z$-transforms are needed and we can work directly with a table that gives right tail areas under the $\chi^2$ distribution. That table is the **Chi-squared Distribution Table**, in the Appendix, and it gives values of $\chi^2$ for given values of area to the right of $\chi^2$, see Figure 8.11.

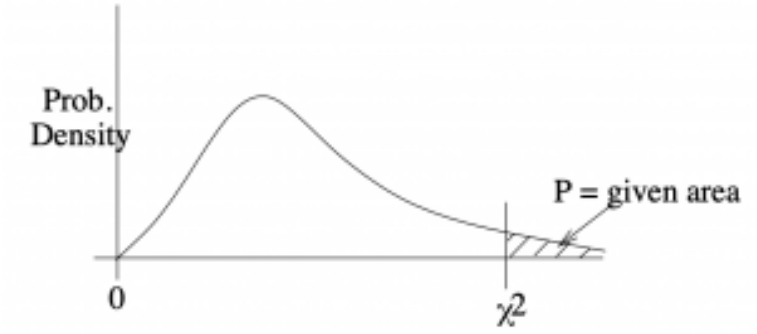



*Figure 8.11 : The Chi-squared Distribution Table gives $\chi^2$ associated with given right tail areas.*

We'll need $\chi^2_{\text{left}}$ and $\chi^2_{\text{right}}$ such that the tail areas are equal and such that the area between them is $C$, see Figure 8.12.

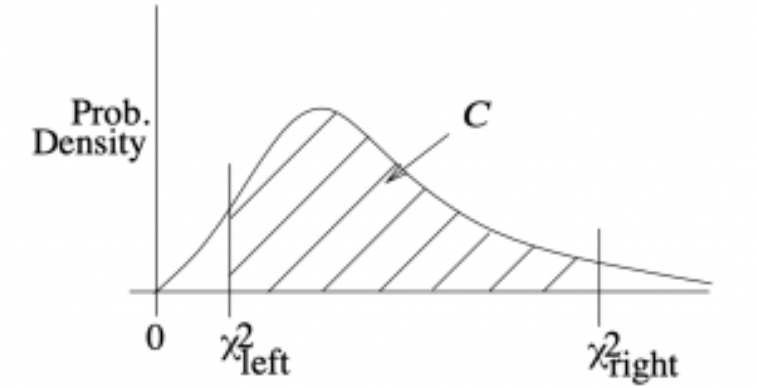



*Figure 8.12 : The values* $\chi^2_{\text{left}}$ *and* $\chi^2_{\text{right}}$ *define the confidence region* $\mathcal{C}$.

*Notation* : Let's call the $\alpha$ in the **Chi-squared Distribution Table** $\alpha_T$ and let $\chi^2(\alpha_T)$ be the table value that corresponds to $\alpha_T$. In other words $\chi^2(\alpha_T)$ is the $\chi^2$ value that corresponds to a right tail area of $\alpha_T$.

So given $\mathcal{C}$, the appropriate $\chi^2_{\text{left}}$ and $\chi^2_{\text{right}}$ are the following values from the **Chi-squared Distribution Table**:

$$\chi^2_{\text{right}} = \chi^2\left(\frac{1-\mathcal{C}}{2}\right)$$

$$\chi^2_{\text{left}} = \chi^2\left(1-\left[\frac{1-\mathcal{C}}{2}\right]\right).$$

Note the symmetry of the **Chi-squared Distribution Table**. If $\chi^2_{\text{right}}$ comes from the column 3 columns from the right edge of the table then $\chi^2_{\text{left}}$ comes from a column 3 columns from the left edge of the table. Only small and large areas appear in the table, there are no intermediate values.

Finally, the confidence interval for $\sigma^2$ is given by

$$\frac{(n-1)s^2}{\chi^2_{\text{right}}} < \sigma^2 < \frac{(n-1)s^2}{\chi^{-2}_{\text{left}}}$$

and for $\sigma$ by:

$$\sqrt{\frac{(n-1)s^2}{\chi^2_{\text{right}}}} < \sigma < \sqrt{\frac{(n-1)s^2}{\chi^2_{\text{left}}}}$$

Where the $\chi^2$ distribution with $\nu = n - 1$ degrees of freedom (giving the line to use in the **Chi-squared Distribution Table**) is used.

**Example 8.5** : Find the $90\%$ confidence interval on $\sigma$ and $\sigma^2$ for the following data

$$59, 54, 53, 52, 51, 39, 49, 46, 49, 48$$

*Solution* : Compute, using your calculator :

$$s^2 = 28.2$$
$$\nu = n - 1 = 9.$$

From the **Chi-squared Distribution Table**, in the $\nu = 9$ line, find :

$$\chi^2_{\text{right}} = \chi^2\left(\frac{1-0.90}{2}\right) = \chi^2(0.05) = 16.919$$

and

$$\chi^2_{\text{left}} = \chi^2(1-0.05) = \chi^2(0.95) = 3.325$$

So

$$\frac{(n-1)s^2}{\chi^2_{\text{right}}} < \sigma^2 < \frac{(n-1)s^2}{\chi^2_{\text{left}}}$$

$$\frac{9 \cdot 28.2}{16.919} < \sigma^2 < \frac{9 \cdot 28.2}{3.325}$$

$$15.0 < \sigma^2 < 76.3 \qquad \text{with } 90\% \text{ confidence.}$$

Taking square roots:

$$3.87 < \sigma < 8.73 \qquad \text{with } 90\% \text{ confidence.}$$

□